

RESEARCH PRESENTATION

DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference
Optimization

arXiv:2504.02882

Multi-Turn DPO for TA-LLMs



PROBLEM

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs trained only with SFT struggle with incomplete queries and out-of-scope requests, leading to hallucinated tool calls.



Failing to ask follow-up questions — Model skips slot-filling and attempts tool call with missing arguments



Hallucinated tool invocations — Model invents argument values instead of asking the user



Failed tool call rejection — Model accepts requests for unavailable tools instead of gracefully declining

5-State Markov Decision Process for TA-LLM Dialogue

Internal states model the dialogue progression from user query to tool response.



States 1-3

Pre-execution: tool selection and slot-filling via QA

States 4-5

Post-execution: await results and generate completion

State 1 (Reject)

Out-of-scope requests return rejection, stay at Initial

Three Query Types Define Distinct State Trajectories

Type	State Trajectory	Description	Example
Type 1	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed.	"Book flight from NYC to LA on May 15"
Type 2	1 → (2xN) → 3 → 4 → 5	Query lacks arguments. Slot-filling QA repeats N times until all fields gathered.	"Book a flight" → "Where from?" → "NYC" → "Where to?" → "LA"
Type 3	1 → 1	Requested functionality unavailable. Tool call must be rejected.	"Call my dentist" (no dentist tool available)

METHOD

Automatic Paired Trajectory Construction

Chosen (correct) and rejected (incorrect) trajectories paired automatically — no human labeling.

Query	Chosen	Rejected	Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	● Easy
Type 1	1→3→4→5	1→(2xN)→3→4→5	Prevent redundant slot-filling	562	● Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	● Easy
Type 2	1→(2xN)→3→4→5	1→(2xM)→3→4→5	Prevent slot hallucination	562	● Hard
Type 3	1→1	1→3→4	Tool call reject	567	● Hard

RESULTS

SFT + DiaTool-DPO Achieves Best Macro Average

FunctionChat-Bench evaluation across three model sizes.

Model	Method	Call	Comp.	Slot	Rel.	Micro	Macro
Prop. 8B	DiaTool-DPO	0.314	0.700	0.833	0.609	0.575	0.614
	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop. 3.1B	DiaTool-DPO	0.357	0.551	0.528	0.391	0.455	0.457
	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO	0.029	0.449	0.056	0.261	0.205	0.199
	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

KEY RESULT

LLaMA-3-8B: Massive Gains with DiaTool-DPO

Slot accuracy and relevance improve dramatically when adding DiaTool-DPO to SFT.



Key Takeaways

1

MDP formulation enables systematic trajectory pairing

5 internal states + 3 query types → automatic chosen/rejected pairs without human annotation

2

SFT + DiaTool-DPO consistently outperforms

Best Macro Avg across all model sizes (3.1B, 8B, LLaMA-3-8B)

3

Slot-filling and tool rejection see largest gains

LLaMA-3-8B: +43.5% Slot accuracy, +10.5% Relevance over SFT-only

4

DPO complements, not replaces, supervised training

DPO-alone performs poorly; DPO + SFT is the winning combination